5.1 p. 271 #5*, 7*, 13, 17, 19, 21, 23

5. $\begin{bmatrix} -4 & 3 & 3 \\ 2 & -3 & -2 \\ -1 & 0 & -2 \end{bmatrix}\begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -12-6+3 \\ 6+6-2 \\ -3+0-2 \end{bmatrix} = \begin{bmatrix} -15 \\ 10 \\ -5 \end{bmatrix}$ $\quad \bar{x} = -5x$ so yes

$A \qquad x \qquad \bar{x} \qquad \lambda = -5$

7. $\overset{A}{\begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 1 \\ -3 & 4 & 5 \end{bmatrix}} - \lambda I = \begin{bmatrix} -1 & 0 & -1 \\ 2 & -1 & 1 \\ -3 & 4 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 2 & -1 & 1 \\ -3 & 4 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 4 & 4 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $A-\lambda I$ is (nontriv sol) consistent so yes

$x_1 = -x_3$

$x_2 = -x_3$ $\quad \cdot x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$ if $x_3 \neq 0$

$\begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$

13. $\lambda = 1$: $\begin{bmatrix} 3 & 0 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 3 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 3 & 0 & 1 \\ 0 & 0 & -\frac{1}{3} \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1/3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$x_1 = -\frac{1}{3}x_3$ $\quad \cdot x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$x_3 = 0$ $\quad$ so $\quad$ basis $= \{e_2\}$

$\lambda = 2$: $\begin{bmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ -2 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$ $\quad x_1 = -\frac{1}{2}x_3$, $\; x_2 = x_3$

basis: $\left\{\begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}\right\}$

$\lambda = 3$: $\begin{bmatrix} 1 & 0 & 1 \\ -2 & -2 & 0 \\ -2 & 0 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & -2 & 2 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$ $\quad x_1 = -x_3$, $\; x_2 = x_3$

basis: $\left\{\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}\right\}$

17. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & -2 \end{bmatrix}$ $\lambda = 0, 3, -2$

19. $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$ $\lambda = 0$ b/c col. lin. dep. so not invertible

21. a) False must be non trivial
    b) True
    c) True
    d) True
    e) False get eigen vectors not values

23. if n+1 eigenvalues, then n+1 eigenvectors that are lin. indep.
but in $\mathbb{R}^n$ so n+1 vectors can't be lin. indep.

5.2 p. 579 #5, 7, 11, 18, 21, 25a, 27a

5. $\begin{bmatrix} 8 & 4 \\ 4 & 8 \end{bmatrix} - \lambda I = \begin{bmatrix} 8-\lambda & 4 \\ 4 & 8-\lambda \end{bmatrix}$ $\det \begin{bmatrix} 8-\lambda & 4 \\ 4 & 8-\lambda \end{bmatrix} = 0$

a) $(8-\lambda)^2 - 16 = 0 \rightarrow (\lambda-12)(\lambda-4)=0$

$8-\lambda = \pm 4$

b) $\lambda = 12, 4$

7. $\begin{vmatrix} 5-\lambda & 3 \\ -4 & 4-\lambda \end{vmatrix} = (5-\lambda)(4-\lambda)+12 = 0$

$20 - 9\lambda + \lambda^2 + 12 = 0$

$\lambda^2 - 9\lambda + 32 = 0$

$\lambda = \frac{9 \pm \sqrt{81-4(32)}}{2}$

no real eigen values

11. $\begin{vmatrix} 3-\lambda & 0 & 0 \\ 2 & 1-\lambda & 4 \\ 1 & 0 & 4-\lambda \end{vmatrix} = (3-\lambda)\begin{vmatrix} 1-\lambda & 4 \\ 0 & 4-\lambda \end{vmatrix} - 0 + 0 = 0$

$(3-\lambda)(1-\lambda)(4-\lambda) = 0$

$-\lambda^3 + 8\lambda^2 - 19\lambda + 12 = 0$

18. $A = \begin{bmatrix} 4 & 2 & 3 & 3 \\ 0 & 2 & h & 3 \\ 0 & 0 & 4 & 14 \\ 0 & 0 & 0 & 2 \end{bmatrix}$ $A - 4I = \begin{bmatrix} 0 & 2 & 3 & 3 \\ 0 & -2 & h & 3 \\ 0 & 0 & 0 & 14 \\ 0 & 0 & 0 & -2 \end{bmatrix} \sim \begin{bmatrix} 0 & 2 & 3 & 0 \\ 0 & 0 & h+3 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

need 2 free variables so $h+3=0$ or $h=-3$

21. a) False only if diagonal

b) False only row replacement doesn't

c) True

d) False $\lambda = -5$

25a. from 4.9 $Av_1 = v_1$ so v_1 eigenvector w/ $\lambda = 1$

$$\begin{vmatrix} .6-\lambda & .3 \\ .4 & .7-\lambda \end{vmatrix} = (.6-\lambda)(.7-\lambda) - .12 = 0$$

$$\lambda^2 - 1.3\lambda + .3 = 0 \qquad \lambda = 1, .3$$

$$(\lambda - 1)(\lambda - .3) = 0$$

$$\begin{bmatrix} .3 & .3 \\ .4 & .4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{so } v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \qquad \text{basis } \mathbb{R}^2 \text{ is } \{v_1, v_2\}$$

27a. $Av_1 = v_1 \quad Av_2 = .5v_2 \quad Av_3 = .2v_3$

5.3 p.286 #1, 3*, 5*, 9, 11, 15, 21

1. $P=\begin{bmatrix}5&7\\2&3\end{bmatrix}$, $D=\begin{bmatrix}2&0\\0&1\end{bmatrix}$ $A^4=\begin{bmatrix}5&7\\2&3\end{bmatrix}\begin{bmatrix}2^4&0\\0&1^4\end{bmatrix}\begin{bmatrix}3&-7\\-2&5\end{bmatrix}\cdot\frac{1}{15-14}$

$=\begin{bmatrix}5&7\\2&3\end{bmatrix}\begin{bmatrix}16&0\\0&1\end{bmatrix}\begin{bmatrix}3&-7\\-2&5\end{bmatrix}=\begin{bmatrix}80&7\\32&3\end{bmatrix}\begin{bmatrix}3&-7\\-2&5\end{bmatrix}=\begin{bmatrix}226&-525\\90&-209\end{bmatrix}$

3. $\begin{bmatrix}a&0\\2(a-b)&b\end{bmatrix}^k=\begin{bmatrix}1&0\\2&1\end{bmatrix}\begin{bmatrix}a^k&0\\0&b^k\end{bmatrix}\begin{bmatrix}1&0\\-2&1\end{bmatrix}$

$A^k=\begin{bmatrix}a^k&0\\2a^k&b^k\end{bmatrix}\begin{bmatrix}1&0\\-2&1\end{bmatrix}=\begin{bmatrix}a^k&0\\2a^k-2b^k&b^k\end{bmatrix}=\begin{bmatrix}a^k&0\\2(a^k-b^k)&b^k\end{bmatrix}$

5. $\begin{bmatrix}2&-1&-1\\1&4&1\\-1&-1&2\end{bmatrix}=\begin{bmatrix}1&-1&0\\-1&1&-1\\0&-1&1\end{bmatrix}\begin{bmatrix}3&0&0\\0&2&0\\0&0&3\end{bmatrix}\begin{bmatrix}0&-1&-1\\-1&-1&-1\\-1&-1&0\end{bmatrix}$

A = P D P^{-1}

$\lambda=3$: $\left\{\begin{bmatrix}1\\-1\\0\end{bmatrix}\begin{bmatrix}0\\-1\\1\end{bmatrix}\right\}$

$\lambda=2$: $\left\{\begin{bmatrix}-1\\1\\-1\end{bmatrix}\right\}$

9. $\det(A-\lambda I)=\begin{vmatrix}2-\lambda&-1\\1&4-\lambda\end{vmatrix}=(2-\lambda)(4-\lambda)+1=0\rightarrow\lambda^2-6\lambda+9=0$

$(\lambda-3)^2=0$

$\lambda=3$: $\begin{bmatrix}-1&-1\\1&1\end{bmatrix}\sim\begin{bmatrix}1&1\\0&0\end{bmatrix}\therefore\begin{bmatrix}-1\\1\end{bmatrix}$

not possible to diagonalize

11. $\begin{bmatrix}0&1&1\\2&1&2\\3&3&2\end{bmatrix}=\begin{bmatrix}-1&-1&1\\1&0&2\\0&1&3\end{bmatrix}\begin{bmatrix}-1&0&0\\0&-1&0\\0&0&5\end{bmatrix}\begin{bmatrix}-1/3&2/3&-1/3\\-1/2&-1/2&1/2\\1/6&1/6&1/6\end{bmatrix}$

A = P D P^{-1}

$\lambda=-1$: $\begin{bmatrix}1&1&1\\2&2&2\\3&3&3\end{bmatrix}\sim\begin{bmatrix}1&1&1\\0&0&0\\0&0&0\end{bmatrix}$

$x_1=-x_2-x_3$ $\left\{\begin{bmatrix}-1\\1\\0\end{bmatrix},\begin{bmatrix}-1\\0\\1\end{bmatrix}\right\}$

$\lambda=5$: $\begin{bmatrix}-5&1&1\\2&-4&2\\3&3&-3\end{bmatrix}\sim\begin{bmatrix}1&0&-1/3\\0&1&-2/3\\0&0&0\end{bmatrix}$

$x_1=1/3\,x_3$

$x_2=2/3\,x_3$

$\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}\right\}$

15. $\begin{bmatrix} 0 & -1 & -1 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ -1 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

$A \qquad P \qquad D \qquad P^{-1}$

$\lambda = 0$: $\begin{bmatrix} 0 & -1 & -1 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $x_1 = x_3$, $x_2 = -x_3$ $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

$\lambda = 1$: $\begin{bmatrix} -1 & -1 & -1 \\ 1 & 1 & 1 \\ -1 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $x_1 = -x_2 - x_3$ $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

$[P \; I] = \begin{bmatrix} -1 & -1 & 1 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 0 & 0 & 1 & 2 & 1 \\ 0 & 1 & 0 & -1 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix} = [I \; P^{-1}]$.

21. a) false (D doesn't have to be diagonal)
b) true
c) false (n eigenvectors)
d) false (invertibility depends on 0 not being an eigenvalue)

5.4 p.293 #1, 3, 5, 11, 15, 17, 20, 31

1. $T(b_1) = 3d_1 - 5d_2 \quad T(b_2) = -d_1 + 6d_2 \quad T(b_3) = 4d_2$

$$\begin{bmatrix} 3 & -1 & 0 \\ -5 & 6 & 4 \end{bmatrix}$$

3. a) $T(e_1) = 0b_1 - 0b_2 + b_3$
$T(e_2) = -b_1 - 2b_2 + 0b_3$
$T(e_3) = 2b_1 + 0b_2 + 3b_3$

$T(x_1, x_2, x_3) = (2x_3 - x_2)b_1 - 2x_2b_2 + (x_1 + 3x_3)b_3$

$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad e = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

b) $[T(e_1)]_B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad [T(e_2)]_B = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix} \quad [T(e_3)]_B = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$

c) $\begin{bmatrix} 0 & -1 & 2 \\ 0 & -2 & 0 \\ 1 & 0 & 3 \end{bmatrix}$

5. a) $(t+3)(3 - 2t + t^2) = 3t - 2t^2 + t^3 + 9 - 6t + 3t^2 = 9 - 3t + t^2 + t^3$

b) p, q poly, c scalar

$T(p+q) = (t+3)(p+q) = (t+3)p + (t+3)q$
$= T(p) + T(q)$

$T(cp) = (t+3)(cp) = c(t+3)p = cT(p)$

c) $B = \{b_1, b_2, b_3\} \quad C = \{c_1, c_2, c_3, c_4\}$
$\quad\quad 1 \quad t \quad t^2 \quad\quad\quad 1 \quad t \quad t^2 \quad t^3$

$T(p) = (t+3)p$

$T(b_1) = T(1) = (t+3)(1) = 3 + t \quad [T(b_1)]_C = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

$T(b_2) = T(t) = (t+3)(t) = 3t + t^2 \quad [T(b_2)]_C = \begin{bmatrix} 0 \\ 3 \\ 1 \\ 0 \end{bmatrix}$

$T(b_3) = T(t^2) = (t+3)(t^2) = 3t^2 + t^3 \quad [T(b_3)]_C = \begin{bmatrix} 0 \\ 0 \\ 3 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} 3 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

11. $P = [b_1 \; b_2] = \begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix}$

B-matrix $= P^{-1}AP = \frac{1}{-1+2}\begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix}\begin{bmatrix} -4 & -1 \\ 6 & 1 \end{bmatrix}\begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -2 & -2 \\ 0 & -1 \end{bmatrix}$

$A = PDP^{-1}$

B = basis for $\mathbb{R}^n$ $\quad B = \{P_1$
from col. of P
D is B-matrix of
$T: x \mapsto Ax$

15. $A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$ $\quad \det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 2 \\ 3 & -4-\lambda \end{vmatrix} = (1-\lambda)(-4-\lambda) - 6 = 0$

$\lambda^2 + 3\lambda - 10 = 0$

$(\lambda+5)(\lambda-2) = 0 \quad \lambda = 2, -5$

$\lambda = 2$: $\begin{bmatrix} -1 & 2 \\ 3 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \therefore v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ $\quad \lambda = -5$: $\begin{bmatrix} 6 & 2 \\ 3 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1/3 \\ 0 & 0 \end{bmatrix} \therefore v_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

$P = [v_1\ v_2]$ which diagonalizes A

Theorem 8, $B = \{v_1, v_2\}$ has property of B-matrix is a diagonal matrix

17. a) $Ab_1 = \begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \end{bmatrix} = 3b_1$ $\quad \begin{vmatrix} 4-\lambda & 1 \\ -1 & 2-\lambda \end{vmatrix} = (4-\lambda)(2-\lambda) + 1 = 0$

$\lambda^2 - 6\lambda + 9 = 0$

only eigenvalue $3 = \lambda$ w/ multiplicity 2 $\quad (\lambda-3)^2 = 0$

$\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ $\quad v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ only 1 eigenvector

$\therefore$ not diagonalizable

$P = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$

b) $P = [b_1\ b_2]$

$B\text{matrix} = P^{-1}AP = \frac{1}{2-1}\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 4 \\ 3 & 3 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$

20. If A similar to B, then A^2 similar to B^2

$A = PBP^{-1} \rightarrow A^2 = (PBP^{-1})(PBP^{-1}) = PB^2P^{-1}$

31. $A = \begin{bmatrix} -7 & -48 & -16 \\ 1 & 14 & 6 \\ -3 & -45 & -19 \end{bmatrix}$ $\quad b_1 = \begin{bmatrix} -3 \\ 1 \\ -3 \end{bmatrix}$ $\quad b_2 = \begin{bmatrix} -2 \\ 1 \\ -3 \end{bmatrix}$ $\quad b_3 = \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix}$

$P = [b_1\ b_2\ b_3] = \begin{bmatrix} -3 & -2 & 3 \\ 1 & 1 & -1 \\ -3 & -3 & 0 \end{bmatrix}$

$B\text{ matrix} = P^{-1}AP = \begin{bmatrix} -1 & -3 & -1/3 \\ 1 & 3 & 0 \\ 0 & -1 & -1/3 \end{bmatrix}\begin{bmatrix} -7 & -48 & -16 \\ 1 & 14 & 6 \\ -3 & -45 & -19 \end{bmatrix}\begin{bmatrix} -3 & -2 & 3 \\ 1 & 1 & -1 \\ 3 & -3 & 0 \end{bmatrix}$

$= \begin{bmatrix} -7 & -2 & -6 \\ 0 & -4 & -6 \\ 0 & 0 & -1 \end{bmatrix}$